From Snapshots to Streams: A Live State Graph Approach for LLM-Based Root-Cause-Analysis in Kubernetes

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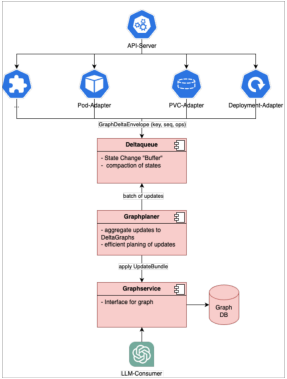

Outline

1 Introduction

2 Background

Introduction

- Kubernetes and cloud-native technologies have become industry standard
- Key challenges: perceived complexity, lack of training [1]
- Root cause analysis (RCA) as key bottleneck

$\Rightarrow$ LLM-based assistance systems promise cognitive reduction and timely detection

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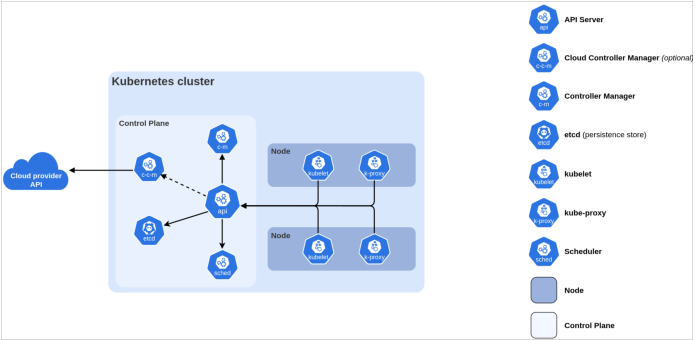
Kubernetes:

- Open-source, feature-rich container orchestrator
- Deployments through declarative manifests



Background

Error sources:

- Unforeseen events: node dies, network issues, ...
- Syntactically correct but semantically incorrect manifests

References I

Cloud Native Computing Foundation.
Cloud native 2024: Approaching a decade of code, cloud, and change, 2024.
[Online].